

Find Largest Element in array:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int largest = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] > largest) {
                largest = arr[i];
            }
        }

        System.out.println(largest);
    }
}
```

Find Smallest Element In Array:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int smallest = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] < smallest) {
                smallest = arr[i];
            }
        }

        System.out.println(smallest);

        sc.close();
    }
}
```

Find First Element In array:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.println(arr[0]);

        sc.close();
    }
}
```

Find Last Element in array

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int n = sc.nextInt();  
        int[] arr = new int[n];  
  
        for (int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }  
  
        System.out.println(arr[n - 1]);  
  
        sc.close();  
    }  
}
```

Largest Element in 2 array:

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int n = sc.nextInt();  
  
        int[] arr1 = new int[n];  
        int[] arr2 = new int[n];  
        int[] result = new int[n];  
  
        for (int i = 0; i < n; i++) {  
            arr1[i] = sc.nextInt();  
        }  
  
        for (int i = 0; i < n; i++) {  
            arr2[i] = sc.nextInt();  
        }  
  
        for (int i = 0; i < n; i++) {  
            result[i] = arr1[i] + arr2[i];  
        }  
  
        int largest = result[0];  
  
        for (int i = 1; i < n; i++) {  
            if (result[i] > largest) {  
                largest = result[i];  
            }  
        }  
    }  
}
```

```
    }  
}  
  
System.out.println(largest);  
  
sc.close();  
}  
}
```

Reverse an array:

Type 1:

```
import java.util.Scanner;  
  
public class Main {  
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

int n = sc.nextInt();
int[] arr = new int[n];

for (int i = 0; i < n; i++) {
    arr[i] = sc.nextInt();
}

for (int i = n - 1; i >= 0; i--) {
    System.out.print(arr[i] + " ");
}

sc.close();
}
}
```

Type 2:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
int n = sc.nextInt();
int[] arr = new int[n];

for (int i = 0; i < n; i++) {
    arr[i] = sc.nextInt();
}

int start = 0;
int end = n - 1;

while (start < end) {
    int temp = arr[start];
    arr[start] = arr[end];
    arr[end] = temp;

    start++;
    end--;
}

for (int num : arr) {
    System.out.print(num + " ");
}

sc.close();
}
```


Linear search:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int key = sc.nextInt();

        int index = -1;

        for (int i = 0; i < n; i++) {
            if (arr[i] == key) {
                index = i;
                break;
            }
        }

        System.out.println(index);
    }
}
```

```
        sc.close();
    }
}
```

Frequency of elements:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];
        boolean[] visited = new boolean[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        for (int i = 0; i < n; i++) {
            if (visited[i]) {
                continue;
            }

            int count = 1;
```

```

        for (int j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                count++;
                visited[j] = true;
            }
        }

        System.out.println(arr[i] + " -> " + count);
    }

    sc.close();
}

```

Remove duplicates:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];
    }
}

```

```

for(int i = 0; i < n; i++) {
    arr[i] = sc.nextInt();
}

for(int i = 0; i < n; i++) {

    boolean isDuplicate = false;

    for(int j = 0; j < i; j++) {
        if(arr[i] == arr[j]) {
            isDuplicate = true;
            break;
        }
    }

    if(!isDuplicate) {
        System.out.print(arr[i] + " ");
    }
}

sc.close();
}
}

```

Program :

```
import java.util.Scanner;
```

```
public class Main {
```

```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int n = sc.nextInt();
    int[] arr = new int[n];

    int sum = 0;

    for (int i = 0; i < n; i++) {
        arr[i] = sc.nextInt();

        if (arr[i] > 0) {
            sum += arr[i];
        }
    }

    System.out.println("The sum of the positive numbers in the array is " + sum);

    sc.close();
}
}

```

Program 8:

```

import java.util.Scanner;

public class Main {

```

```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int n = sc.nextInt();
    int[] arr = new int[n];

    int sum = 0;

    for (int i = 0; i < n; i++) {
        arr[i] = sc.nextInt();

        if (arr[i] % 2 == 0) {
            sum += arr[i];
        }
    }

    System.out.println("The sum of the even numbers in the array is " + sum);

    sc.close();
}
}

```

Program 9:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

```

```
Scanner sc = new Scanner(System.in);
```

```
int n = sc.nextInt();
```

```
int[] arr1 = new int[n];
```

```
int[] arr2 = new int[n];
```

```
for (int i = 0; i < n; i++) {  
    arr1[i] = sc.nextInt();  
}
```

```
for (int i = 0; i < n; i++) {  
    arr2[i] = sc.nextInt();  
}
```

```
boolean compatible = true;
```

```
for (int i = 0; i < n; i++) {  
    if (arr1[i] < arr2[i]) {  
        compatible = false;  
        break;  
    }  
}
```

```
if (compatible) {  
    System.out.println("Compatible");  
}
```

```
    } else {  
        System.out.println("Incompatible");  
    }  
  
    sc.close();  
}  
}
```